Regime-switching RL model for Portfolio Optimization

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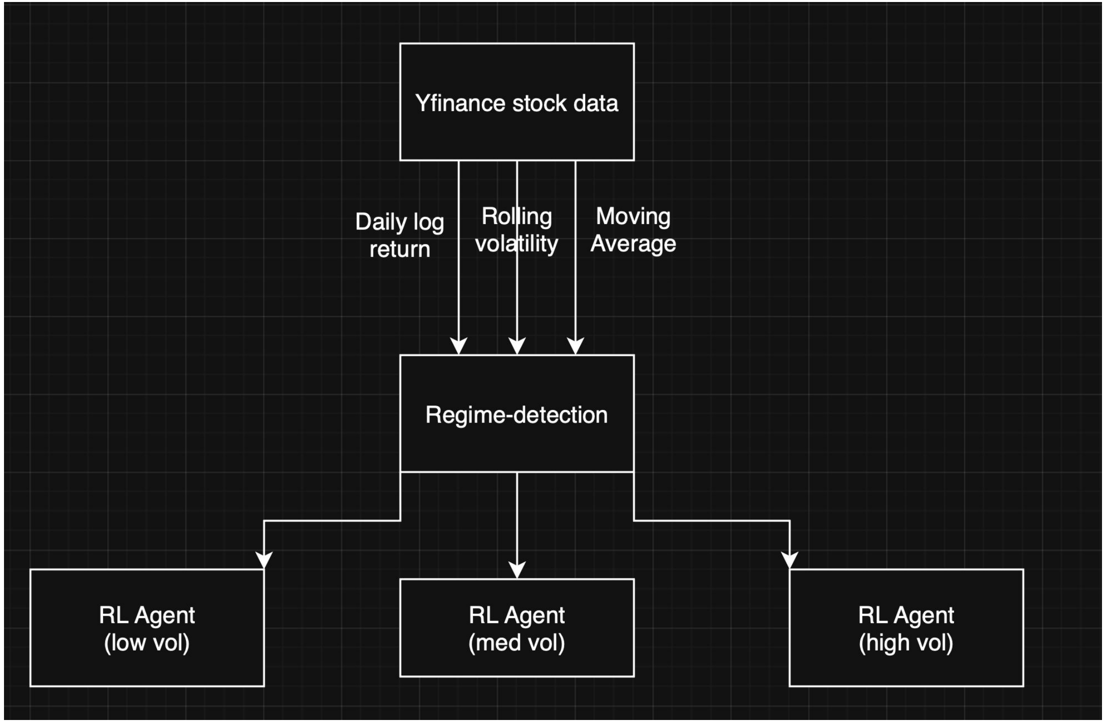
High-level Architecture



Initial Simple regime detection

1. Data Engineering Pipeline

- S&P 500 Index daily OHLCV from `yfinance`.
- Calculates **20-Day Rolling Volatility** of daily log returns to ensure statistical stationarity.
- Log returns are used to ensure the data is suitable for statistical modeling

2. The Classification Logic

The system categorizes market states into three distinct buckets based on historical relative volatility:

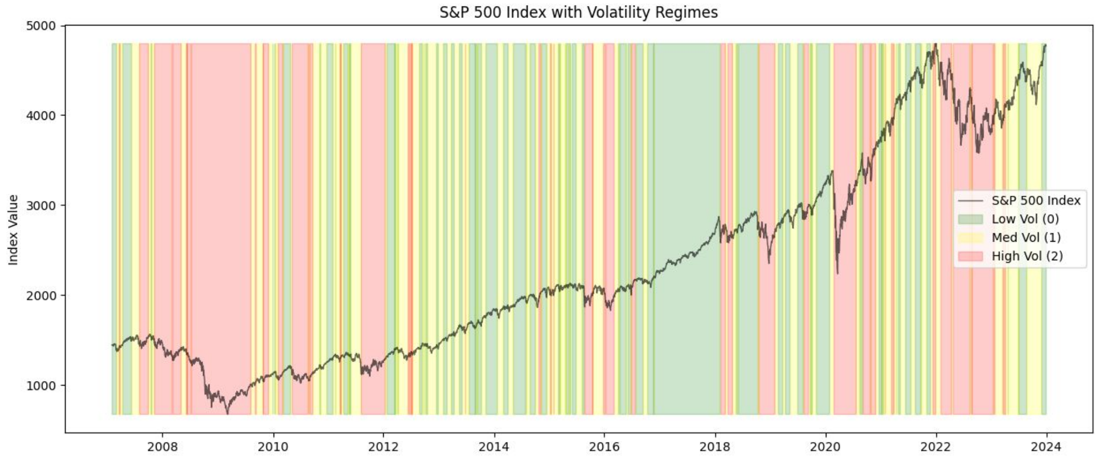
- **Low Vol (Regime 0):** Volatility < 33rd percentile
- **Med Vol (Regime 1):** Volatility between 33rd and 67th percentile
- **High Vol (Regime 2):** Volatility > 67th percentile

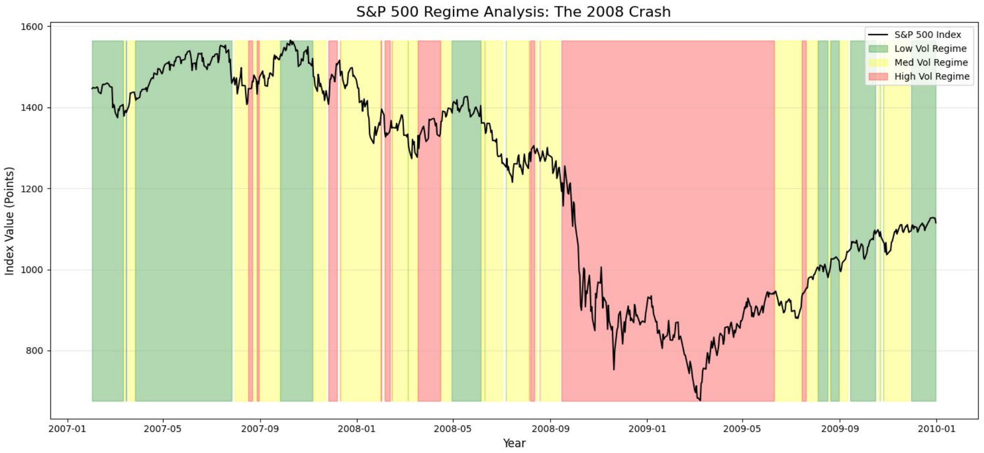
Log returns -

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Volatility of daily log return -

$$\sigma_t = \sqrt{\frac{1}{N-1}\sum_{i=0}^{N-1}(r_{t-i} - \bar{r})^2}$$


S&P 500 Index with Volatility Regimes
Index Value
S&P 500 Index
Low Vol (0)
Med Vol (1)
High Vol (2)


S&P 500 Regime Analysis: The 2008 Crash
Index Value (Points)
Year
S&P 500 Index
Low Vol Regime
Med Vol Regime
High Vol Regime

Plan for next week

- **HMM Model Development**
    - Define transition states, probabilities,
- **Implement baseline from paper**
    - Check different sectors volatility
- Implement base architecture/pipeline
    -

Baseline: DRL vs MVO Implementation

- Reproduced Paper 3: PPO agent vs Mean-Variance Optimization
- Universe: 11 US sector ETFs (XLB–XLRE), 2006–2024
- RL agent used - PPO, reward function Differential Sharpe Ratio reward
- MVO: Ledoit-Wolf shrinkage, max-Sharpe objective
- Walk-forward sliding-window evaluation

Baseline: DRL vs MVO Implementation

Metric	DRL	MVO
Sharpe Ratio	1.17	0.68
Annualized Return	12.11%	6.53%
Max Drawdown	-32.96%	-33.03%

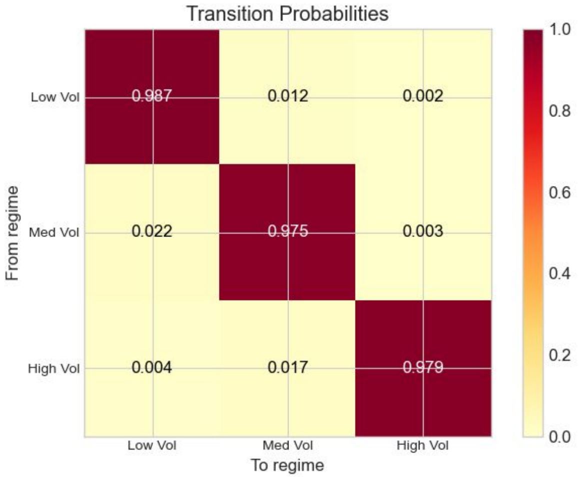
Limitation: Single agent — no regime awareness, same strategy in across all volatility regimes.

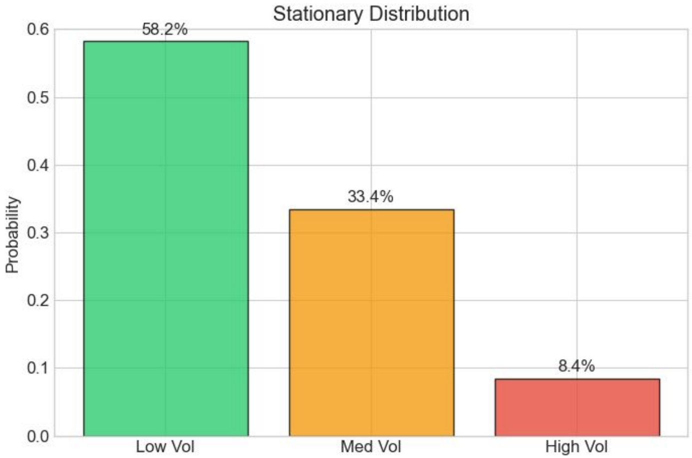
HMM Regime Detection

- Gaussian HMM, 3 hidden states, full covariance, EM algorithm (hmmlearn)
- 3 features: daily log return, 20-day rolling vol, MA ratio (20/60)
- Train: S&P 500 2006–2019 (3,522 days) | Test: 2020–2024 (1,258 days)
- Model selection: tested n = 2, 3, 4, 5 → 3 regimes optimal (BIC = −72,105, test LL = 12,530)
- Transition matrix shows high persistence (regimes are "sticky")

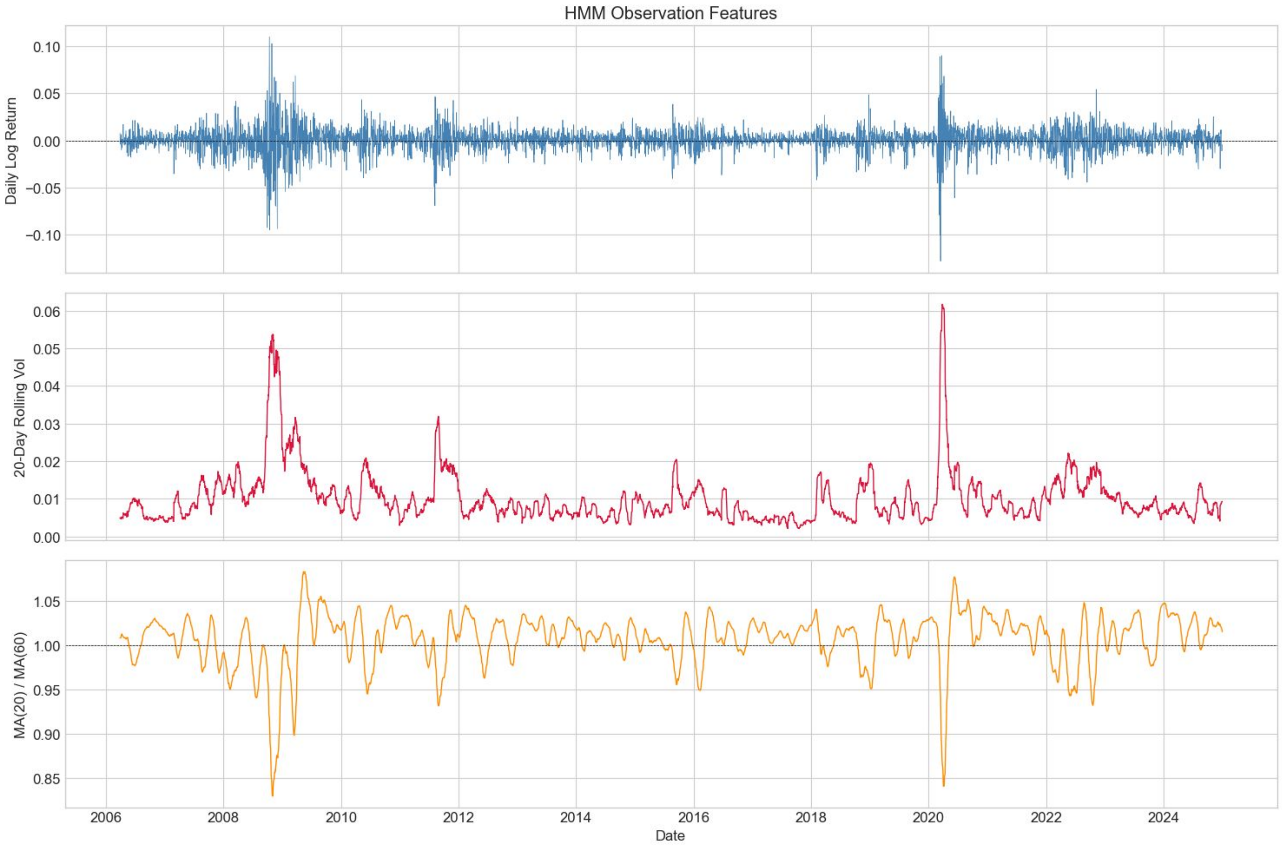
HMM Regime Detection

- Used EM algorithm
  - Objective function is maximize the Log likelihood





HMM Regime Detection

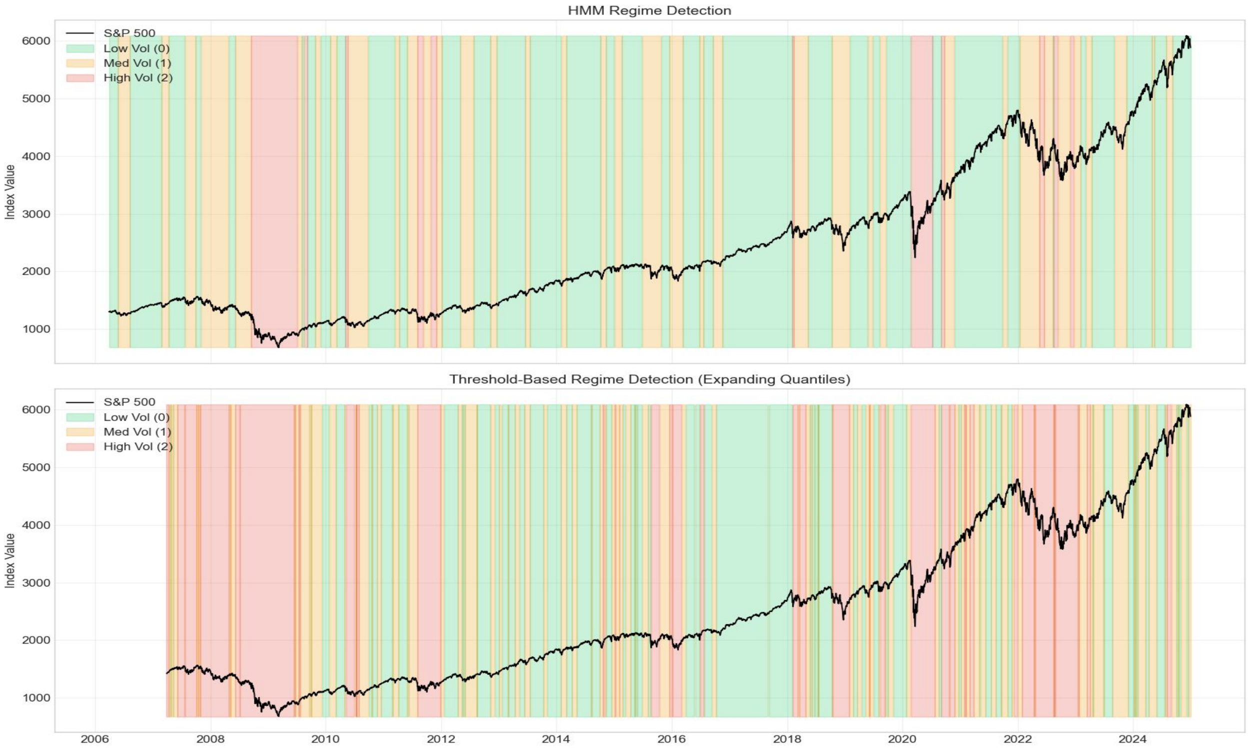


HMM Regime Detection - Why 3 regimes?

- Adding more regimes basically increasing overfitting.
- 3 regimes has the lowest BIC (Bayesian information criterion)

HMM model

- Figure showing the difference b/w HMM and threshold based regime detection

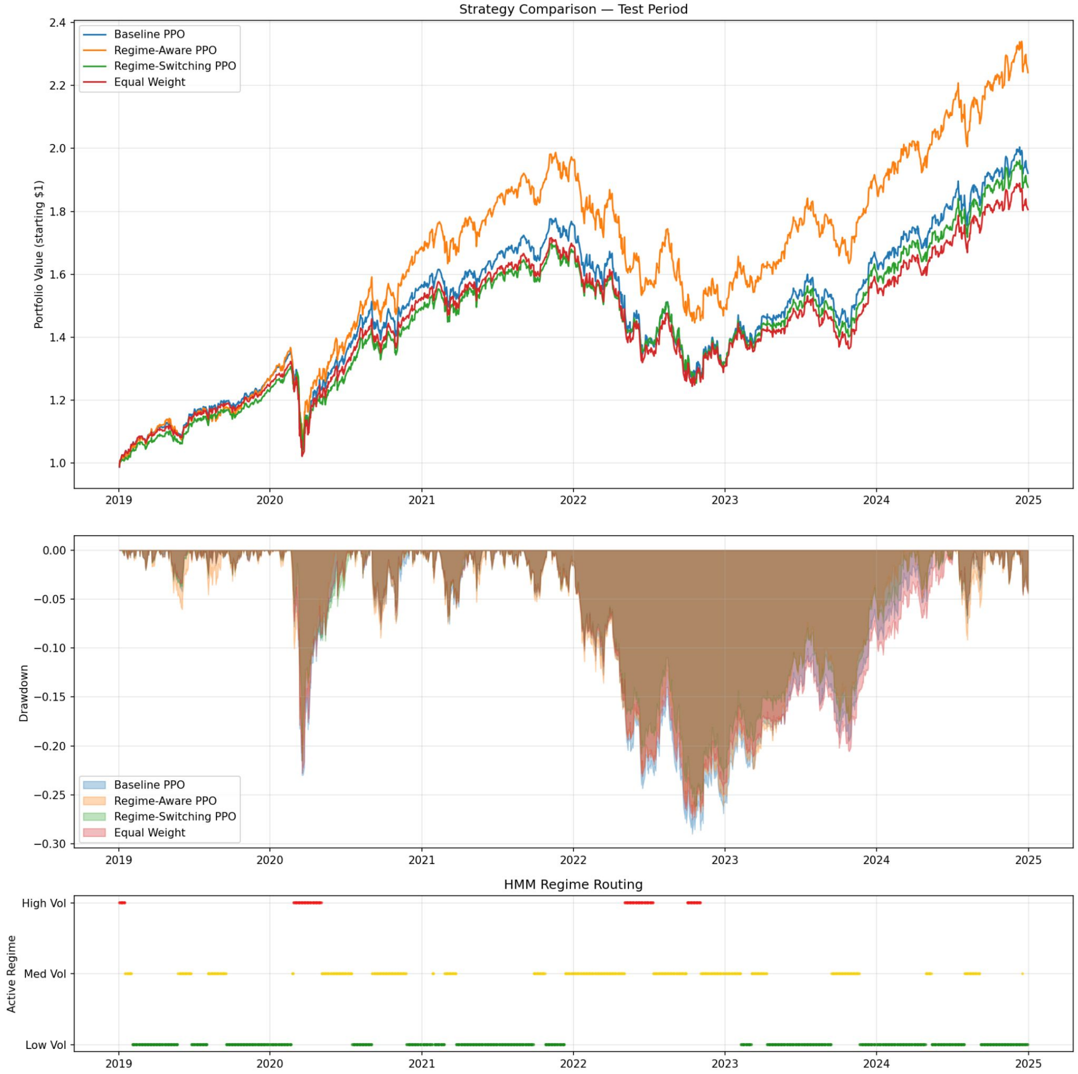


Next steps

- Train 3 specialized PPO agents — one per regime (low / med / high vol)
- Regime-switching evaluation: HMM routes market state → matching agent takes action
- Compare regime-switching system vs single generalist agent (baseline)
- **Case studies for different regimes compared to prev. implementations.**
- **Novel Design choice of HMM.**

Baseline implementation overview

- Train 3 specialized PPO agents — one per regime (low / med / high vol)
- Regime-switching evaluation: HMM routes market state → matching agent takes action


Strategy Comparison — Test Period
Baseline PPO
Regime-Aware PPO
Regime-Switching PPO
Equal Weight
Portfolio Value (starting $1)
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
2019
2020
2021
2022
2023
2024
2025
Drawdown
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
Baseline PPO
Regime-Aware PPO
Regime-Switching PPO
Equal Weight
HMM Regime Routing
Active Regime
High Vol
Med Vol
Low Vol

Comparison against existing methods

Following is the comparison of the model against previous model (Sood et. al)

Both models trained for 500k timesteps and uses 9 sector ETFs.

Strategy	Sharpe	Max Drawdown	Volatility	Cumulative Return
Regime-Switching PPO	1.17	-16.1%	11.2%	54.3%
Baseline PPO	0.82	-40.2%	15.7%	31.8%
MVO	0.67	-28.5%	13.9%	24.5%
Equal-Weight	0.59	-32.7%	14.1%	21.2%

COVID case study (Feb - March 2020)

- HMM detected the high-volatility early, triggering the high vol. agent

Strategy	Return	Volatility	Max Drawdown
Regime-Switching PPO	-6.6%	12.2%	-6.9%
Baseline PPO	-29.4%	60.9%	-29.6%
MVO	-28.6%	85.3%	-35.5%
Equal Weight	-33.5%	70.5%	-33.7%

Next Steps

- Hysteresis based routing
- Try to account qualitative data like news, user sentiment etc. to determine volatility.
- Check for switching latency and impacts